Algebraic hyperbolicity and line-containing degree ranges in rational homogeneous varieties (Theorem 1.4)

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Editorial note. Entire Theorem 1.4 = Theorem 4.3 is one claim, not two ranges.

Editorial note. Native mathematical source is unmodified; journal style and standalone logo are excluded.

Editorial scope. Author-material preparation only; no mathematical repair or formal admission.

Allowed standard prerequisites (editorial).

- Vertical tangent and normal bundles, genus identity, and section-dominating surjection. Original Proposition 2.1, Lemma 2.2, Proposition 2.7; Coskun–Riedl [CR23], DOI 10.1007/s11856-022-2379-2, corresponding Proposition 2.1, Lemma 2.2, Proposition 2.7. Statements and definitions retained in Section 2; their external proofs are not present in this source and are not claimed reproduced.
- Rank-growth and rank-one quotient scroll. Clemens [Cle03] Section 2.2, DOI 10.1081/agb-120022438; original Lemma 3.4 citing Coskun–Riedl [CR23] Lemma 2.13. Allowed earlier tools; scroll lemma statement retained but its external proof is not reproduced.
- Deformation of lines and homogeneous geometry. Debarre [Deb01] Theorem 2.6; Ramanathan [Ram87] Theorem 3.11/Remark 3.12; Manivel [Man20] Theorem 1.5 and following paragraph; Kobayashi–Ochiai [KO73]. Cited deformation dimension, quadratic equations, covering by lines and canonical-index bounds; author applications and computations retained.
- Chow rings, Schubert classes and Chern classes. Eisenbud–Harris [EH16] Proposition 4.9, Theorem 6.34, Proposition 6.4, Chapter 5; Liu [Liu22] Proposition 3.1. Pieri, line-scheme dimension/class, Whitney and splitting principles are standard referenced tools. Unlike these external tools, the full author proof of Lemma 4.1 is retained.
- Contextual homogeneous embeddings and previously known examples. Ramanan–Ramanathan [R-R85]; Coskun–Riedl [CR23]; Yeong [Yeo25] Lemma 4.2. Supporting examples and comparison/open range only; no independent counting of contextual examples or conjecture.

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Scope and changes. One target: the entire Theorem 1.4, restated as Theorem 4.3, with both degree ranges. The definition of algebraic hyperbolicity, Main Setting 1.3, and complete Sections 2–4 are retained verbatim from the fixed native source. The section numbering is restored; journal furniture is replaced by a standard wrapper. Section 5 applications and unrelated introductory discussion are not selected. No mathematical argument has been supplied or repaired.

Dependency boundary. Cited earlier results are allowed prerequisites; their statements and original references are retained, but their external proofs are not represented as reproduced. In particular this applies to the Coskun–Riedl normal-bundle and scroll tools. The retained author proofs include the new quantitative cases and the full Schubert argument. The final conjecture and illustrative examples are context, not counted results or asserted proofs.

Preparation only. No formal problem number, mathematical certification, or final global uniqueness is asserted. Later in-flight overlap after the saved 14 October 2026 cutoff is unconfirmed. AI assistance was used only for selection, metadata and mechanical packaging; no author AI-use assertion is inferred.

A complex projective variety X is *algebraically hyperbolic* if, for some ample divisor H, there exists a real number $\epsilon > 0$ such that, for any integral curve $C \subset X$, the inequality

$$2g(C) - 2 \geq \epsilon \deg_H(C)$$

is satisfied, where $g(C)$ is the geometric genus of C. The goal of this paper is to prove the algebraic hyperbolicity of very general hypersurfaces in homogeneous varieties by generalizing techniques developed by Clemens [Cle86, Cle03], Ein [Ein88, Ein91], Voisin [Voi96, Voi98], Pacienza [Pac03, Pac04], Clemens and Ran [CR04], Coskun and Riedl [CR19, CR23], and Yeong [Yeo25].

Let us start by defining the general setting we will refer to throughout the paper.

Main Setting 1.3. Let A be a smooth complex projective variety such that

- A is a rational homogeneous variety with a transitive action by an algebraic group G;
- A has an embedding into a product of projective spaces

  $$i\colon A \hookrightarrow \mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$$

  such that the divisors $H_1, \ldots, H_m$ of A corresponding to the pullbacks of the hyperplane classes $\mathcal{O}_{\mathbb{P}^{N_i}}(1)$ generate the Picard group of A;
- the embedding i is projectively normal; that is, the restriction

  $$H^0\left(\mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}, \mathcal{O}_{\mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}}(d_1, \ldots, d_m)\right) \longrightarrow H^0(A, d_1H_1 + \cdots + d_mH_m)$$

  is surjective for all $d_1, \ldots, d_m > 0$.

Write the class of the canonical divisor of A as $K_A = a_1H_1 + \cdots + a_mH_m$. Let $\mathcal{E}$ be the line bundle $d_1H_1 + \cdots + d_mH_m$ for $d_1, \ldots, d_m > 0$, equivariant under G, and assume that $H_1, \ldots, H_m$ is a collection of section-dominating line bundles for $\mathcal{E}$ (see Definition 2.3). A degree $(d_1, \ldots, d_m)$ hypersurface X of A is a section of $\mathcal{E}$.

Theorem 1.4. *Let A be a rational homogeneous variety as in Main Setting 1.3.*

- *If $\dim A \geq 4$ and $d_i \geq \dim A - a_i - 2$ for all $1 \leq i \leq m$, then a very general hypersurface X of degree $(d_1, \ldots, d_m)$ is algebraically hyperbolic.*
- *If $d_i \leq \dim A - a_i - 4$ for some $1 \leq i \leq m$, then a general hypersurface X of degree $(d_1, \ldots, d_m)$ contains lines. In particular, X is not algebraically hyperbolic.*

The cases with $d_i \geq \dim A - a_i - 3$ for all $1 \leq i \leq m$ and $d_i = \dim A - a_i - 3$ for some $1 \leq i \leq m$ remain open. When $\dim A \leq 4$, there are examples when it fails to imply algebraic hyperbolicity (see Examples 4.4 and 4.5.) For $A = \mathbb{P}^4$, the case of sextic threefolds remains open. However, when $\dim A \geq 5$, it does imply algebraic hyperbolicity for $A = \mathbb{P}^n$ and $A = \mathbb{P}^m \times \mathbb{P}^n$; see [Yeo25]. We conjecture X is algebraically hyperbolic in these cases when $\dim A \geq 5$ (see Conjecture 4.6.)

We then apply the theorem to concrete examples of homogeneous varieties. We note that the theorem can also be applied to products of these varieties.

2 Preliminaries

2.1 Setup

We first describe the general setup for the techniques developed by Clemens and Ein [Ein88, Ein91], Voisin [Voi96, Voi98], Pacienza [Pac03, Pac04], Coskun and Riedl [CR19, CR23], and Yeong [Yeo25].

Let A be a smooth, complex projective variety of dimension D, and assume A admits a transitive group action by an algebraic group G. Let $\mathcal{E}$ a globally generated vector bundle on A, equivariant under G, of rank $r < D - 1$, and let $V = H^0(A, \mathcal{E})$. We will be particularly interested in the case when $\mathcal{E}$ is a very ample line bundle on A.

Let X be the zero locus of a very general section of $\mathcal{E}$, and suppose X contains a curve Y of degree e and geometric genus g. Then, if $\mathcal{X}_1$ is the universal hypersurface over V, we get the relative Hilbert scheme $\mathcal{H} \to V$ with universal curve $\mathcal{Y}_1$, where the general fiber of $\mathcal{Y}_1 \to \mathcal{H}$ is a curve of geometric genus g and degree e. We can find a G-invariant subvariety $U \subset \mathcal{H}$ such that the map $U \to V$ is étale. Restricting $\mathcal{Y}_1$ to U and taking a resolution of the general fiber, we get a smooth family $\mathcal{Y} \to U$ whose fibers are

smooth curves of genus g. We pull $\mathcal{X}_1$ back to a family $\mathcal{X}$ over U, with projection maps $\pi_1\colon \mathcal{X} \to U$ and $\pi_2\colon \mathcal{X} \to A$. There is a natural generically injective map $h\colon \mathcal{Y} \to \mathcal{X}$.

We define the *vertical tangent sheaf* $T_{\mathcal{X}/A}$ by the short exact sequence

$$0 \longrightarrow T_{\mathcal{X}/A} \longrightarrow T_{\mathcal{X}} \longrightarrow \pi_2^* T_A \longrightarrow 0.$$

Since we constructed $\mathcal{Y}$ to be stable under the G-action, we have that $\pi_2 \circ h$ dominates A and that the map $T_{\mathcal{Y}} \to h^*\pi_2^* T_A$ is surjective. Similarly, we define the vertical tangent sheaf $T_{\mathcal{Y}/A}$ as the kernel in the short exact sequence

$$0 \longrightarrow T_{\mathcal{Y}/A} \longrightarrow T_{\mathcal{Y}} \longrightarrow h^*\pi_2^* T_A \longrightarrow 0.$$

Let $M_{\mathcal{E}}$ denote the *Lazarsfeld–Mukai bundle* associated to $\mathcal{E}$, defined by the sequence

$$0 \longrightarrow M_{\mathcal{E}} \longrightarrow H^0(A, \mathcal{E}) \otimes \mathcal{O}_A \longrightarrow \mathcal{E} \longrightarrow 0.$$

For a general element t of U, let Y_t be the fiber of $\mathcal{Y}$ over t and X_t be the fiber of $\mathcal{X}$ over t. Denote by $h_t\colon Y_t \to X_t$ the restriction of h to Y_t. The following result translates the relation between the vertical tangent sheaves and the Lazarsfeld–Mukai bundle.

Proposition 2.1 (*cf.* [CR23, Proposition 2.1])**.**

(1) $N_{h_t/X_t} \cong N_{h/\mathcal{X}}|_{Y_t}$.

(2) $T_{\mathcal{X}/A} \cong \pi_2^* M_{\mathcal{E}}$.

(3) $N_{h/\mathcal{X}}$ *is the cokernel of the map of vertical tangent sheaves* $T_{\mathcal{Y}/A} \to h^* T_{\mathcal{X}/A}$.

2.2 The degree of the normal bundle and section-dominating line bundles

There is a close relation between the degree of the normal bundle N_{h_t/X_t} and the genus of the curve Y_t described by the following lemma. We can use it to bound the genus of Y_t by finding a bound for the degree of the normal bundle N_{h_t/X_t}.

Lemma 2.2 (*cf.* [CR23, Lemma 2.2])**.** *The following holds:*

$$\deg\left(N_{h_t/X_t}\right) = 2g(Y_t) - 2 - K_{X_t} \cdot h_t(Y_t).$$

By Proposition 2.1, we have a surjection

$$M_{\mathcal{E}}|_{Y_t} \longrightarrow N_{h_t/X_t}. \tag{2.1}$$

We can get a surjection to N_{h_t/X_t} from a simpler bundle by using the notion of *section-dominating* line bundles.

Definition 2.3 (*cf.* [CR23, Definition 2.3])**.** Let $\mathcal{E}$ be a vector bundle on A. A collection of non-trivial globally generated line bundles $L_1, \ldots, L_m$ is a *section-dominating collection* of line bundles for $\mathcal{E}$ if $\mathcal{E} \otimes L_i^\vee$ is globally generated for every $1 \le i \le m$ and the map

$$\bigoplus_{i=1}^{m} \left(H^0(L_i \otimes \mathcal{I}_p) \otimes H^0\left(\mathcal{E} \otimes L_i^\vee\right)\right) \longrightarrow H^0(\mathcal{E} \otimes \mathcal{I}_p)$$

is surjective for every point $p \in A$.

Example 2.4. Suppose $A = \mathbb{P}^n$, and let H be the hyperplane class. Then, for any $\mathcal{E} = dH$ with $d > 0$, H is a section-dominating line bundle for $\mathcal{E}$. To prove this, we need to show that

$$H^0\left(\mathcal{O}_{\mathbb{P}^n}(1) \otimes \mathcal{I}_p\right) \otimes H^0\left(\mathcal{O}_{\mathbb{P}^n}(d-1)\right) \longrightarrow H^0\left(\mathcal{O}_{\mathbb{P}^n}(d) \otimes \mathcal{I}_p\right)$$

is surjective. Choose coordinates $x_0, \ldots, x_n$ for $\mathbb{P}^n$ so that $p = (1 : 0 : \cdots : 0)$. Then $H^0(\mathcal{O}_{\mathbb{P}^n}(d) \otimes \mathcal{I}_p)$ is the set of degree d forms without x_0^d term, and these can be written as a combination of products of linear forms without x_0 term in $H^0(\mathcal{O}_{\mathbb{P}^n}(1) \otimes \mathcal{I}_p)$ and degree $d-1$ forms in $H^0(\mathcal{O}_{\mathbb{P}^n}(d-1))$.

Example 2.5. Let $A = G(k, n)$ be the Grassmannians of k-planes in the affine n-space, and let H correspond to the hyperplane class, that is, the pullback $p^*\mathcal{O}_{\mathbb{P}^N}(1)$ via the Plücker embedding $p\colon G(k,n) \to \mathbb{P}^N$. Then, for any $\mathcal{E} = dH$ with $d > 0$, H is a section-dominating line bundle for $\mathcal{E}$. This follows directly from the previous example and the projective normality of the Grassmannian under the Plücker embedding (*cf.* [RR85].) By [RR85], a similar result is true for flag varieties and Schubert varieties.

Example 2.6. This property can be extended to products: Let $A = \prod_{i=1}^m A_i$ be a product of varieties A_i such that, for each i, H_i is a section-dominating line bundle for $\mathcal{E}_i = d_i H_i$ for some $d_i > 0$. Then $H_1, \dots, H_m$ is a section-dominating collection of line bundles for $\mathcal{E} = \sum_{i=1}^m d_i H_i$.

Given a section of $H^0(\mathcal{E} \otimes L_i^\vee)$, the natural multiplication map defines maps $L_i \to \mathcal{E}$ and $H^0(L_i) \to H^0(\mathcal{E})$, and induces a map $M_{L_i} \to M_{\mathcal{E}}$. By choosing bases for $H^0(\mathcal{E} \otimes L_i^\vee)$, we can get a surjection to $M_{\mathcal{E}}$.

Proposition 2.7 (*cf.* [CR23, Proposition 2.7])**.** *Let $\mathcal{E}$ be a globally generated vector bundle and $M_{\mathcal{E}}$ the Lazarsfeld–Mukai bundle associated to $\mathcal{E}$. Let $L_1, \dots, L_m$ be a section-dominating collection of line bundles for $\mathcal{E}$. Then, there is a surjection*

$$\bigoplus_{i=1}^m M_{L_i}^{\oplus s_i} \longrightarrow M_{\mathcal{E}}$$

for some integers s_i.

From Proposition 2.7 and the surjection (2.1), we obtain a surjection to the normal bundle

$$\beta\colon \bigoplus_{i=1}^m M_{L_i}^{\oplus s_i}|_{Y_t} \longrightarrow N_{h_t/X_t}. \tag{2.2}$$

By choosing generic sections of $H^0(\mathcal{E} \otimes L_i^\vee)$ for the multiplication in Proposition 2.7, the rank of the image of β is a strictly increasing function of the s_i until surjectivity is reached (*cf.* [Cle03, Section 2.2].) In particular, we can choose a surjection β so that $s_1 + \cdots + s_m \leq \operatorname{rk} N_{h_t/X_t}$.

This surjection can be used to get a bound for the degree of N_{h_t/X_t} in terms of the integers s_i.

Lemma 2.8. *Let $\beta\colon \bigoplus_{i=1}^m M_{L_i}^{\oplus s_i}|_{Y_t} \to N_{h_t/X_t}$ be a surjective map as in* (2.2)*. Then*

$$\deg N_{h_t/X_t} \geq -\sum_{i=1}^m s_i \deg L_i|_{Y_t}.$$

Proof. Let K be the kernel of β:

$$0 \longrightarrow K \longrightarrow \bigoplus_{i=1}^m M_{L_i}^{\oplus s_i}|_{Y_t} \xrightarrow{\beta} N_{h_t/X_t} \longrightarrow 0.$$

Then by the definition of the M_{L_i}, we get an injection $K \to \bigoplus_{i=1}^m H^0(A, L_i)^{\oplus s_i} \otimes \mathcal{O}_A|_{Y_t}$. Since it is a direct sum of trivial bundles, $\bigoplus_{i=1}^m H^0(A, L_i)^{\oplus s_i} \otimes \mathcal{O}_A|_{Y_t}$ is a semi-stable vector bundle. It follows that

$$\frac{\deg K}{\operatorname{rk} K} = \mu(K) \leq 0,$$

so

$$\deg(K) \leq 0;$$

hence, from the sequence above and the sequence defining M_{L_i}, we have

$$\deg N_{h_t/X_t} = \deg\left(\bigoplus_{i=1}^m M_{L_i}^{\oplus s_i}|_{Y_t}\right) - \deg K \geq \deg\left(\bigoplus_{i=1}^m M_{L_i}^{\oplus s_i}|_{Y_t}\right) = -\sum_{i=1}^m s_i \deg L_i|_{Y_t}. \qquad \square$$

By combining Lemmas 2.2 and 2.8, we obtain a bound for the genus:

$$2g(Y_t) - 2 \geq K_{X_t} \cdot Y_t - \sum_{i=1}^m s_i \deg L_i|_{Y_t}. \tag{2.3}$$

Therefore, if we can control the integers s_i, we can prove the algebraic hyperbolicity of X_t. This leads us to our first criterion for hyperbolicity. This result is similar to [CR23, Corollary 2.9] and [HI21, Theorem 3.6].

Proposition 2.9. *Let A be a homogeneous variety of dimension D, and let $\mathcal{E}$ be a very ample line bundle on A as in Setup 2.1. Let $L_1, \dots, L_m$ be a collection of section-dominating line bundles for $\mathcal{E}$ on A. If, for some $\epsilon > 0$,*

$$K_{X_t} \cdot Y_t \geq (D - 2 + \epsilon) \cdot \sum_{i=1}^m \deg L_i|_{Y_t},$$

then X_t is algebraically hyperbolic.

Proof. As before, we can choose the s_i so that $s_1 + \cdots + s_m \leq \operatorname{rk} N_{h_t/X_t} = D - 2$. In particular, we can assume $s_i \leq D - 2$ for every $1 \leq i \leq m$. Thus, by Inequality (2.3), we get

$$2g(Y_t) - 2 \geq K_{X_t} \cdot Y_t - (D-2) \sum_{i=1}^{m} \deg L_i|_{Y_t}.$$

Therefore, if $K_{X_t} \cdot Y_t \geq (D - 2 + \epsilon) \cdot \sum_{i=1}^{m} \deg L_i|_{Y_t}$, we get

$$2g(Y_t) - 2 \geq \epsilon \cdot \sum_{i=1}^{m} \deg L_i|_{Y_t}.$$

Hence, X_t is algebraically hyperbolic. □

Remark 2.10. This initial setup can also be defined when A is not homogeneous but contains a Zariski-open homogeneous set A_0. Results equivalent to Proposition 2.9 can be obtained in this case; see, for example, [CR23, Theorem 1.2]. This allowed Coskun and Riedl [CR23] to classify algebraically hyperbolic surfaces X in the cases when A is a Hirzebruch surface $\mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(e))$, A is $\mathbb{P}^3$ blown up at a single point, and A is a weighted projective space $\mathbb{P}(1,1,1,n)$. We believe the generalizations made in this paper can also be worked out when A contains a Zariski-open homogeneous set A_0, allowing future studies of higher-dimensional toric varieties, such as blowups of projective spaces and weighted projective spaces.

3 Scroll method for homogeneous varieties

In this section, we work out the scroll method introduced by Coskun and Riedl in [CR23] in a more general setting. This technique builds a surface scroll on the curve Y_t whose degree can be used to improve the bound for algebraic hyperbolicity obtained in Proposition 2.9.

Here we will work on rational homogeneous varieties satisfying the Main Setting 1.3, which we recall here:

Let A be a smooth complex projective variety of dimension D such that

- A is a rational homogeneous variety with a transitive action by an algebraic group G;
- A has a projectively normal embedding into a product of projective spaces $\mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$;
- the divisors $H_1, \ldots, H_m$ of A corresponding to the pullbacks of the hyperplane classes $\mathcal{O}_{\mathbb{P}^{N_i}}(1)$ generate the Picard group of A.

Let $\mathcal{E}$ be the line bundle $d_1H_1 + \cdots + d_mH_m$ for $d_1, \ldots, d_m > 0$, equivariant under G, and assume that $H_1, \ldots, H_m$ is a collection of section-dominating line bundles for $\mathcal{E}$. To simplify the notation, we write X for the zero locus of a very general section of $\mathcal{E}$, and C for the curve in X, instead of X_t and Y_t.

Write the class of the canonical divisor of A as $K_A = a_1H_1 + \cdots + a_mH_m$. By the adjunction formula, $K_X = \sum_{i=1}^{m}(a_i + d_i)H_i|_X$. In this case, Proposition 2.9 can be rephrased as follows.

Corollary 3.1. *If there exists an $\epsilon > 0$ such that $d_i \geq (D - a_i - 2) + \epsilon$ for all $1 \leq i \leq m$, then X is algebraically hyperbolic.*

Proof. Since

$$K_X \cdot C = \sum_{i=1}^{m}(a_i + d_i)H_i \cdot C \geq (D - 2 + \epsilon) \cdot \sum_{i=1}^{m} H_i \cdot C,$$

the result follows from Proposition 2.9. □

The main theorem in this section shows that we do not need the ϵ in the bound above.

Definition 3.2. Given a product of projective spaces $\mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$, the fibers of the projection

$$\pi_1' \colon \mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m} \longrightarrow \mathbb{P}^{N_2} \times \cdots \times \mathbb{P}^{N_m}$$

are $\mathbb{P}^{N_1}$. We define a $\mathbb{P}^{N_1}$-*line* as a line contained in one of these fibers. Equivalently, a $\mathbb{P}^{N_1}$-line is a curve of numerical class $H_1^{N_1-1}H_2^{N_2} \cdots H_m^{N_m}$. We define $\mathbb{P}^{N_i}$-*lines* similarly for $i = 1, \ldots, m$. We call a surface scroll a $\mathbb{P}^{N_i}$-*scroll* if it contains a $\mathbb{P}^{N_i}$-line through every point.

We can see the space of $\mathbb{P}^{N_1}$-lines as the product $G(2, N_1+1) \times \mathbb{P}^{N_2} \times \cdots \times \mathbb{P}^{N_m}$ and the Fano scheme of $\mathbb{P}^{N_1}$-lines contained in A as a subscheme $F_1^{N_1}(A) \subset G(2, N_1+1) \times \mathbb{P}^{N_2} \times \cdots \times \mathbb{P}^{N_m}$, and similarly for $\mathbb{P}^{N_i}$-lines. We will need the dimension of the space of $\mathbb{P}^{N_i}$-lines in A.

Lemma 3.3. *Let $F_1^{N_i}(A)$ be the Fano scheme of $\mathbb{P}^{N_i}$-lines in A. Then*

$$\dim F_1^{N_i}(A) = \dim A - a_i - 3.$$

Proof. Let L be a $\mathbb{P}^{N_i}$-line in A. Since A is a rational homogeneous variety, its tangent bundle T_A is globally generated; thus $T_A|_L$ is globally generated. A classic deformation argument (*cf.* [Deb01, Theorem 2.6]) then shows that $\dim F_1^{N_i}(A) = h^0(L, N_{L/A}) = \dim A - K_A \cdot L - 3 = \dim A - a_i - 3$. □

We use the following lemma to construct a scroll.

Lemma 3.4 (*cf.* [CR23, Lemma 2.13])**.** *A rank 1 quotient Q of $M_{H_i}|_C$ induces a $\mathbb{P}^{N_i}$-scroll Σ over C of H_i-degree equal to* $\deg Q + (H_i \cdot C)$.

Theorem 3.5. *Let A be a rational homogeneous variety as in Main Setting 1.3. If $\dim A \geq 4$ and $d_i \geq \dim A - a_i - 2$ for all $1 \leq i \leq m$, then X is algebraically hyperbolic.*

Proof. By Proposition 2.7, there is a surjection

$$\beta\colon \bigoplus_{i=1}^{m} M_{H_i}^{\oplus s_i}|_C \longrightarrow N_{C/X}.$$

Without loss of generality, assume $s_1 \geq \cdots \geq s_m$. As before, we can assume $s_1 + \cdots + s_m \leq \operatorname{rk} N_{C/X} = D-2$. We divide the proof into the cases $s_1 \leq D-3$ and $s_1 = D-2$.

Case 1. First suppose that $s_1 \leq D-3$. So, $s_i \leq D-3$ for all i. Then, the inequality

$$2g(C) - 2 \geq K_X \cdot C - \sum_{i=1}^{m} s_i \deg H_i|_C$$

from (2.3) implies that

$$2g(C) - 2 \geq \left(\sum_{i=1}^{m}(a_i + d_i)H_i\right) \cdot C - (D-3)\sum_{i=1}^{m}(H_i \cdot C),$$

or equivalently

$$2g(C) - 2 \geq \sum_{i=1}^{m}(a_i + d_i - D + 3)(H_i \cdot C).$$

Since, by hypothesis, $d_i \geq D - a_i - 2$ for $i = 1, \ldots, m$, we get

$$2g(C) - 2 \geq \sum_{i=1}^{m}(H_i \cdot C).$$

Case 2. Now suppose instead that $s_1 = D-2$ is the minimum possible value for s_1 for which β is surjective. So we have $s_2 = \cdots = s_m = 0$ and a surjection

$$\beta\colon M_{H_1}^{\oplus D-2}|_C \longrightarrow N_{C/X}.$$

Since this is the minimum value for s_1 such that β is surjective, this means that the induced map

$$M_{H_1}|_C \longrightarrow N_{C/X}/\beta\left(M_{H_1}^{\oplus D-3}|_C\right) \tag{3.1}$$

has a rank 1 image Q (*cf.* [Cle03, Section 2.2].) By Lemma 3.4, this induces a $\mathbb{P}^{N_1}$-scroll $\Sigma \subset \mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$ containing C of H_1-degree $\Sigma \cdot H_1^2 = \deg Q + (H_1 \cdot C)$. And since Σ is a $\mathbb{P}^{N_1}$-scroll, we have $H_1 H_j \cdot \Sigma = H_j \cdot C$ for $j \neq 1$. Thus, Σ has numerical class

$$(H_1 \cdot C + \deg Q)H_1^{N_1-2}H_2^{N_2}\cdots H_m^{N_m} + (H_2 \cdot C)H_1^{N_1-1}H_2^{N_2-1}\cdots H_m^{N_m} + \cdots + (H_m \cdot C)H_1^{N_1-1}H_2^{N_2}\cdots H_m^{N_m-1}.$$

Since A is projectively normal, there exists a degree $(d_1, \ldots, d_m)$ hypersurface $Z \subset \mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$ such that $X = A \cap Z$. We divide the proof into the cases $\Sigma \subset Z$ and $\Sigma \not\subset Z$.

First suppose that $\Sigma \subset Z$. We will prove in Theorem 4.2 that X does not contain $\mathbb{P}^{N_1}$-lines. Thus, we must have $\Sigma \not\subset A$. As rational homogeneous varieties in projective space are cut out by quadrics (*cf.* [Ram87, Theorem 3.11 and Remark 3.12]), by the Segre embedding, we can express A as cut out by varieties of degree $(2, \dots, 2)$ in $\mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$.

Then, there exists a hypersurface F of class $2H_1 + \cdots + 2H_m$ such that $\Sigma \not\subset F$. Intersecting the classes of Σ and F, we have

$$([\Sigma] \cdot [F]) \cdot H_1 = 2(H_1 \cdot C + \deg Q) + 2(H_2 \cdot C) + \cdots + 2(H_m \cdot C).$$

Since $C \subset \Sigma \cap F$, it follows that $([\Sigma] \cdot [F]) \cdot H_1 \geq H_1 \cdot C$, so

$$2(H_1 \cdot C + \deg Q) + 2(H_2 \cdot C) + \cdots + 2(H_m \cdot C) \geq H_1 \cdot C;$$

hence

$$\deg Q \geq -\frac{1}{2}(H_1 \cdot C) - (H_2 \cdot C) - \cdots - (H_m \cdot C). \tag{3.2}$$

Now, a proof similar to that of Lemma 2.8 for the surjection (3.1) yields the inequality

$$\deg N_{C/X} \geq \deg Q + \deg \beta(M_{H_1}^{\oplus D-3}) \geq \deg Q - (D-3)(H_1 \cdot C).$$

Thus,

$$\deg N_{C/X} = 2g(C) - 2 - (K_X \cdot C) \geq \deg Q - (D-3)(H_1 \cdot C);$$

hence,

$$\begin{aligned} 2g(C) - 2 \geq & (K_X \cdot C) - (D-3)(H_1 \cdot C) + \deg Q \\ = & \sum_{i=1}^{m} (a_i + d_i)(H_i \cdot C) - (D-3)(H_1 \cdot C) + \deg Q, \end{aligned}$$

and by Inequality (3.2), we have

$$2g(C) - 2 \geq \left(a_1 + d_1 - D + 2 + \frac{1}{2}\right)(H_1 \cdot C) + \sum_{i=2}^{m} (a_i + d_i - 1)(H_i \cdot C).$$

Since, by hypothesis, $d_i \geq D - a_i - 2$ for all $1 \leq i \leq m$ and $D \geq 4$, we get

$$2g(C) - 2 \geq \frac{1}{2}(H_1 \cdot C) + \sum_{i=2}^{m} (H_i \cdot C).$$

Therefore, in this case we have $2g(C) - 2 \geq \epsilon \sum_{i=1}^{m} (H_i \cdot C)$ for $\epsilon = \frac{1}{2}$; hence X is algebraically hyperbolic.

Now suppose instead that $\Sigma \not\subset Z$. We intersect Σ and Z to get

$$([\Sigma] \cdot [Z]) \cdot H_1 = d_1(H_1 \cdot C + \deg Q) + d_2(H_2 \cdot C) + \cdots + d_m(H_m \cdot C).$$

As $C \subset \Sigma \cap Z$, we have $([\Sigma] \cdot [Z]) \cdot H_1 \geq H_1 \cdot C$, thus

$$d_1(H_1 \cdot C + \deg Q) + d_2(H_2 \cdot C) + \cdots + d_m(H_m \cdot C) \geq H_1 \cdot C,$$

so

$$\deg Q \geq \left(\frac{1}{d_1} - 1\right)(H_1 \cdot C) - \frac{d_2}{d_1}(H_2 \cdot C) - \cdots - \frac{d_m}{d_1}(H_m \cdot C). \tag{3.3}$$

And as before, we have

$$2g(C) - 2 \geq \sum_{i=1}^{m} (a_i + d_i)(H_i \cdot C) - (D-3)(H_1 \cdot C) + \deg Q,$$

so by Inequality (3.3), we get

$$2g(C) - 2 \geq \left(a_1 + d_1 - D + 2 + \frac{1}{d_1}\right)(H_1 \cdot C) + \sum_{i=2}^{m} \left(a_i + d_i - \frac{d_i}{d_1}\right)(H_i \cdot C).$$

Therefore, it suffices to have $a_1 + d_1 \geq D - 2$ and $a_i + d_i - \frac{d_i}{d_1} > \epsilon$ for some $\epsilon > 0$ and $2 \leq i \leq m$. The first inequality is true by hypothesis. We now check the second inequality.

First, we observe that $a_i \leq -2$ for all $1 \leq i \leq m$. Indeed, if $a_i \geq -1$, then by Lemma 3.3, the dimension of the space of $\mathbb{P}^{N_i}$-lines in A would be $\dim A - a_i - 3 \leq \dim A - 2$, but then the lines could not cover A, which is not the case for a rational homogeneous variety (*cf.* [Man20, Theorem 1.5 and the following paragraph]). And since $d_i \geq D - a_i - 2$, this implies that $d_i \geq 4$ for all $1 \leq i \leq m$.

On the other hand, an n-dimensional projective manifold Y with an ample divisor H satisfies the inequality $K_Y \geq -(n+1)H$, with equality when X is isomorphic to $\mathbb{P}^n$ (*cf.* [KO73].) Thus, looking at each projection $\pi_i \colon A \to \mathbb{P}^{N_i}$, we obtain that $a_i \geq -(D+1)$ for each $1 \leq i \leq m$.

Now, rearrange the inequality we want to prove, $a_i + d_i - \frac{d_i}{d_1} > \epsilon$, as

$$d_i > (\epsilon - a_i)\frac{d_1}{d_1 - 1}.$$

Since $d_1 \geq 4$, we have $\frac{d_1}{d_1 - 1} \leq \frac{4}{3}$, and as $d_i \geq D - a_i - 2$, it suffices to show that

$$D - 2 - a_i > (\epsilon - a_i)\frac{4}{3},$$

or equivalently

$$D - 2 \geq -\frac{a_i}{3} + \epsilon',$$

for $\epsilon' = 4\epsilon/3$.

Now, as $a_i \geq -(D+1)$, we have $-\frac{a_i}{3} + \epsilon' \leq \frac{D+1}{3} + \epsilon'$, so we only need to show that

$$D - 2 \geq \frac{D+1}{3} + \epsilon'.$$

Rearranging terms, this is equivalent to

$$D \geq \frac{7}{2} + \frac{3\epsilon'}{2},$$

which is true since $D \geq 4$.

Therefore, in all cases we get $2g(C) - 2 \geq \epsilon \sum_{i=1}^{m} (H_i \cdot C)$ for some $\epsilon > 0$; hence X is algebraically hyperbolic. □

4 Lines in hypersurfaces of homogeneous varieties

In this section, we obtain the degrees for which the general hypersurface in A contains lines. In particular, we get a bound for when X is not algebraically hyperbolic. First, we collect a lemma characterizing certain Schubert classes in the Grassmannian of lines, based on Liu's work [Liu22].

Lemma 4.1 (*cf.* [Liu22, Proposition 3.1])**.** *Let $d \geq 2$, $m > 0$, and let $X \subset G(2, N)$ be a (possibly reducible) variety with class $[X] = m \cdot \sigma_{N-2,N-2-(d+1)}$ in the Chow ring of the Grassmannian of lines $G(2, N)$. Then, the lines parametrized by X pass through a finite set of fixed points in $\mathbb{P}^{N-1}$.*

Proof. Consider the dual Grassmannian $G(N-2, N)$. The class of the dual variety of $[X] = m \cdot \sigma_{N-2,N-2-(d+1)}$ in $G(N-2, N)$ is $[X^*] = m \cdot \sigma_{2^{N-2-(d+1)},1^{d+1}}$.

By Pieri's formula [EH16, Proposition 4.9], we have $[X^*] \cdot \sigma_2 = 0$. Since σ_2 parametrizes copies of $\mathbb{P}^{N-3}$ through a point in $\mathbb{P}^{N-1}$, it follows that the general point of $\mathbb{P}^{N-1}$ is not contained in any of the spaces $\mathbb{P}^{N-3}$ parametrized by X^*. Hence, the spaces $\mathbb{P}^{N-3}$ parametrized by each component of X^* sweep out a subvariety T of dimension $N-2$ of $\mathbb{P}^{N-1}$. We claim that T is a hyperplane, $T \cong \mathbb{P}^{N-2}$. To see this, consider the universal family

$$\mathcal{U} = \{(x, p) \mid x \in X^*,\ p \text{ lies in the } \mathbb{P}^{N-3} \text{ parametrized by } x\} \subset X^* \times T.$$

The fibers of the projection $\mathcal{U} \to X^*$ are $\mathbb{P}^{N-3}$, so $\dim \mathcal{U} = N - 3 + \dim X^*$. Thus, by the projection $\mathcal{U} \to T$, there is a $(\dim X^* - 1)$-parameter family of copies of $\mathbb{P}^{N-3}$ through each point p of T. Since $d \geq 2$, we have $\dim X^* - 1 \geq 2$, so we get a two-parameter family of copies of $\mathbb{P}^{N-3}$ through each point p of T. Each of these copies of $\mathbb{P}^{N-3}$ is in the tangent space $T_p(T) \cong \mathbb{P}^{N-2}$; thus by dimension, $T_p(T) = \mathbb{P}^{N-2} = T$.

Therefore, there exists one $\mathbb{P}^{N-2} \subset \mathbb{P}^{N-1}$ such that every $\mathbb{P}^{N-3}$ parametrized by a component of X^* is contained in $\mathbb{P}^{N-2}$. Taking back duals, every line parametrized by X passes through a fixed point in $\mathbb{P}^{N-1}$. Taking the union for all components of X, there is a finite set of fixed points in $\mathbb{P}^{N-1}$ such that every line parametrized by X passes through one of these points. □

Theorem 4.2. *Let A be a rational homogeneous variety as in Main Setting 1.3. Let X be a general hypersurface of degree $(d_1, \dots, d_m)$ in A.*

- *If $d_i \leq \dim A - a_i - 4$ for some $1 \leq i \leq m$, then X contains $\mathbb{P}^{N_i}$-lines. In particular, X is not algebraically hyperbolic.*
- *If $d_i > \dim A - a_i - 4$ for some $1 \leq i \leq m$, then X does not contains $\mathbb{P}^{N_i}$-lines.*

Proof. Set $D = \dim A$. To simplify the notation, let us assume that $d_1 \leq D - a_1 - 4$. Then, we want to show that X contains $\mathbb{P}^{N_1}$-lines.

Since A is projectively normal, there is a degree $(d_1, \dots, d_m)$ hypersurface $Z \subset \mathbb{P}^{N_1} \times \cdots \times \mathbb{P}^{N_m}$ such that $X = A \cap Z$. Let $F_1^{N_1}(Z) \subset G(2, N_1 + 1) \times \mathbb{P}^{N_2} \times \cdots \times \mathbb{P}^{N_m}$ be the Fano scheme of $\mathbb{P}^{N_1}$-lines in Z, and let $F_1^{N_1}(A)$ be the Fano scheme of $\mathbb{P}^{N_1}$-lines in A. Denote by $[F_1^{N_1}(Z)]$ and $[F_1^{N_1}(A)]$ their classes in the Chow ring of $G(2, N_1 + 1) \times \mathbb{P}^{N_2} \times \cdots \times \mathbb{P}^{N_m}$. To show that X contains lines, we are going to prove that the intersection class $[F_1^{N_1}(Z)] \cdot [F_1^{N_1}(A)]$ is not zero.

By the definition of degree, a general $\mathbb{P}^{N_1}$ section of Z is a general degree d_1 hypersurface in $\mathbb{P}^{N_1}$. By [EH16, Theorem 6.34], the scheme of lines in a general degree d_1 hypersurface has codimension $d_1 + 1$ in $G(2, N_1 + 1)$. Thus, $F_1^{N_1}(Z)$ has codimension $d_1 + 1$ in $G(2, N_1 + 1) \times \mathbb{P}^{N_2} \times \cdots \times \mathbb{P}^{N_m}$. On the other hand, $\dim F_1^{N_1}(A) = D - a_1 - 3$ by Lemma 3.3. Therefore, it suffices to prove the theorem for $D - a_1 - 3 = d_1 + 1$, or equivalently $d_1 = D - a_1 - 4$.

The intersection of $[F_1^{N_1}(Z)]$ with $H_2^{N_2} \cdots H_m^{N_m}$ is the class of lines in a general $\mathbb{P}^{N_1}$ section of Z, which is a general degree d_1 hypersurface in $\mathbb{P}^{N_1}$. So, by [EH16, Proposition 6.4], we can compute the class $[F_1^{N_1}(Z)]$:

$$\left[F_1^{N_1}(Z)\right] = c_{d_1+1}(\mathrm{Sym}^{d_1} S^*),$$

where S is the universal subbundle of $G(2, N_1 + 1)$.

And since $F_1^{N_1}(A)$ is a space of $\mathbb{P}^{N_1}$-lines, its class $[F_1^{N_1}(A)]$ is also generated by classes of $G(2, N_1 + 1)$.

By the Whitney formula and the splitting principle, see [EH16, Chapter 5], we can write

$$1 + \sigma_1 + \sigma_{1,1} = c(S^*) = (1 + \alpha)(1 + \beta)$$

for $\alpha + \beta = \sigma_1$ and $\alpha\beta = \sigma_{1,1}$. Thus,

$$c(\mathrm{Sym}^{d_1} S^*) = (1 + d_1\alpha)[1 + (d_1 - 1)\alpha + \beta] \cdots [1 + \alpha + (d_1 - 1)\beta](1 + d_1\beta).$$

We are interested in the top Chern class

$$c_{d_1+1}(\mathrm{Sym}^{d_1} S^*) = (d_1\alpha)[(d_1 - 1)\alpha + \beta] \cdots [\alpha + (d_1 - 1)\beta](d_1\beta).$$

Suppose that d_1 is even (the case when d_1 is odd is similar), and rearrange the expression above as

$$\begin{aligned}
&\left[F_1^{N_1}(Z)\right] = c_{d_1+1}(\mathrm{Sym}^{d_1} S^*) \\
&= (d_1\alpha)(d_1\beta)[(d_1 - 1)\alpha + \beta][\alpha + (d_1 - 1)\beta] \cdots [i\alpha + (d_1 - i)\beta][(d_1 - i)\alpha + i\beta] \cdots \left(\frac{d_1}{2}\alpha + \frac{d_1}{2}\beta\right) \\
&= d_1^2(\alpha\beta)[(d_1 - 1)(\alpha + \beta)^2 + (d_1 - 2)^2(\alpha\beta)] \cdots [i(d_1 - i)(\alpha + \beta)^2 + (d_1 - 2i)^2(\alpha\beta)] \cdots \frac{d_1}{2}(\alpha + \beta) \\
&= d_1^2\sigma_{1,1}[(d_1 - 1)\sigma_1^2 + (d_1 - 2)^2\sigma_{1,1}] \cdots [i(d_1 - i)\sigma_1^2 + (d_1 - 2i)^2\sigma_{1,1}] \cdots \frac{d_1}{2}\sigma_1.
\end{aligned}$$

Notice that, after the terms are expanded, all coefficients are non-negative and the term $\sigma_{1,1}\sigma_1^{d_1-1}$ appears. Recall that these Schubert classes multiply by the rule $\sigma_1 \cdot \sigma_{k,l} = \sigma_{k+1,l} + \sigma_{k,l+1}$ and $\sigma_{1,1} \cdot \sigma_{k,l} = \sigma_{k+1,l+1}$. Thus, we can see that, in the expansion of $\sigma_1^{d_1-1}$, all Schubert classes $\sigma_{i,j}$ with $i+j = d_1-1$ appear with positive coefficients. It follows that, in the expansion of $[F_1^{N_1}(Z)]$, all classes $\sigma_{i,j}$ with $i + j = d_1 + 1$ and $i, j \geq 1$ appear with positive coefficients; that is, the only degree $d_1 + 1$ class not in the expansion of $[F_1^{N_1}(Z)]$ is $\sigma_{d_1+1,0}$. Since $\sigma_{N-1,N-1-(d_1+1)}$ is the only Schubert class of dimension $d_1 + 1$ whose intersection with $\sigma_{d_1+1,0}$ in $G(2, N + 1)$ is non-zero, it suffices to show that $[F_1^{N_1}(A)]$ is not a multiple of $\sigma_{N-1,N-1-(d_1+1)}$. But, by Lemma 4.1, if $[F_1^{N_1}(A)]$ is a multiple of $\sigma_{N-1,N-1-(d_1+1)}$, then the lines of a general $\mathbb{P}^{N_1}$ section of A pass through a finite set of points, which cannot happen since A is homogeneous. Therefore, the product $[F_1^{N_1}(Z)] \cdot [F_1^{N_1}(A)]$ is not zero; hence there are $\mathbb{P}^{N_1}$-lines contained in X.

On the other hand, if $d_i > D - a_i - 4$ for some $1 \leq i \leq m$, then the dimension of $F_1^{N_1}(A)$ is less than the codimension of $F_1^{N_1}(Z)$ in $G(2, N_i + 1)$. Since X is general, $F_1^{N_1}(Z)$ is general, and therefore the intersection $F_1^{N_1}(A) \cap F_1^{N_1}(Z)$ is empty. □

By Theorems 3.5 and 4.2, we have our main result.

Theorem 4.3. *Let A be a rational homogeneous variety as in Main Setting 1.3.*

- *If $\dim A \geq 4$ and $d_i \geq \dim A - a_i - 2$ for all $1 \leq i \leq m$, then a very general hypersurface X of degree $(d_1, \ldots, d_m)$ is algebraically hyperbolic.*
- *If $d_i \leq \dim A - a_i - 4$ for some $1 \leq i \leq m$, then a general hypersurface X of degree $(d_1, \ldots, d_m)$ contains $\mathbb{P}^{N_i}$-lines. In particular, X is not algebraically hyperbolic.*

This leaves open the cases with $d_i \geq \dim A - a_i - 3$ for all $1 \leq i \leq m$ and $d_i = \dim A - a_i - 3$ for some $1 \leq i \leq m$. When $\dim A \leq 4$, there are examples when this bound fails to imply algebraic hyperbolicity.

Example 4.4 (*cf.* [Yeo25, Lemma 4.2]). A very general hypersurface $X \subset \mathbb{P}^2 \times \mathbb{P}^2$ of degree (d_1, d_2) with $d_1 = 4$ or $d_2 = 4$ contains an elliptic curve. In particular, it is not algebraically hyperbolic.

The same proof also works for $\mathbb{P}^2 \times \mathbb{P}^1 \times \mathbb{P}^1$.

Example 4.5. A very general hypersurface $X \subset \mathbb{P}^2 \times \mathbb{P}^1 \times \mathbb{P}^1$ of degree (d_1, d_2, d_3) with $d_1 = 4$ contains an elliptic curve. In particular, it is not algebraically hyperbolic.

The bound $d_i \geq \dim A - a_i - 3$ for all i also fails to imply algebraic hyperbolicity in the cases $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2 \times \mathbb{P}^1$, as proved in [CR23]. For $A = \mathbb{P}^4$, the case of sextic threefolds remains open. However, when $\dim A \geq 5$, the bound does imply algebraic hyperbolicity in the cases $A = \mathbb{P}^n$ and $A = \mathbb{P}^m \times \mathbb{P}^n$, as proved in [Yeo25]. We conjecture the same holds for rational homogeneous varieties.

Conjecture 4.6. *If $\dim A \geq 5$ and $d_i \geq \dim A - a_i - 3$ for all i, then is X algebraically hyperbolic.*

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